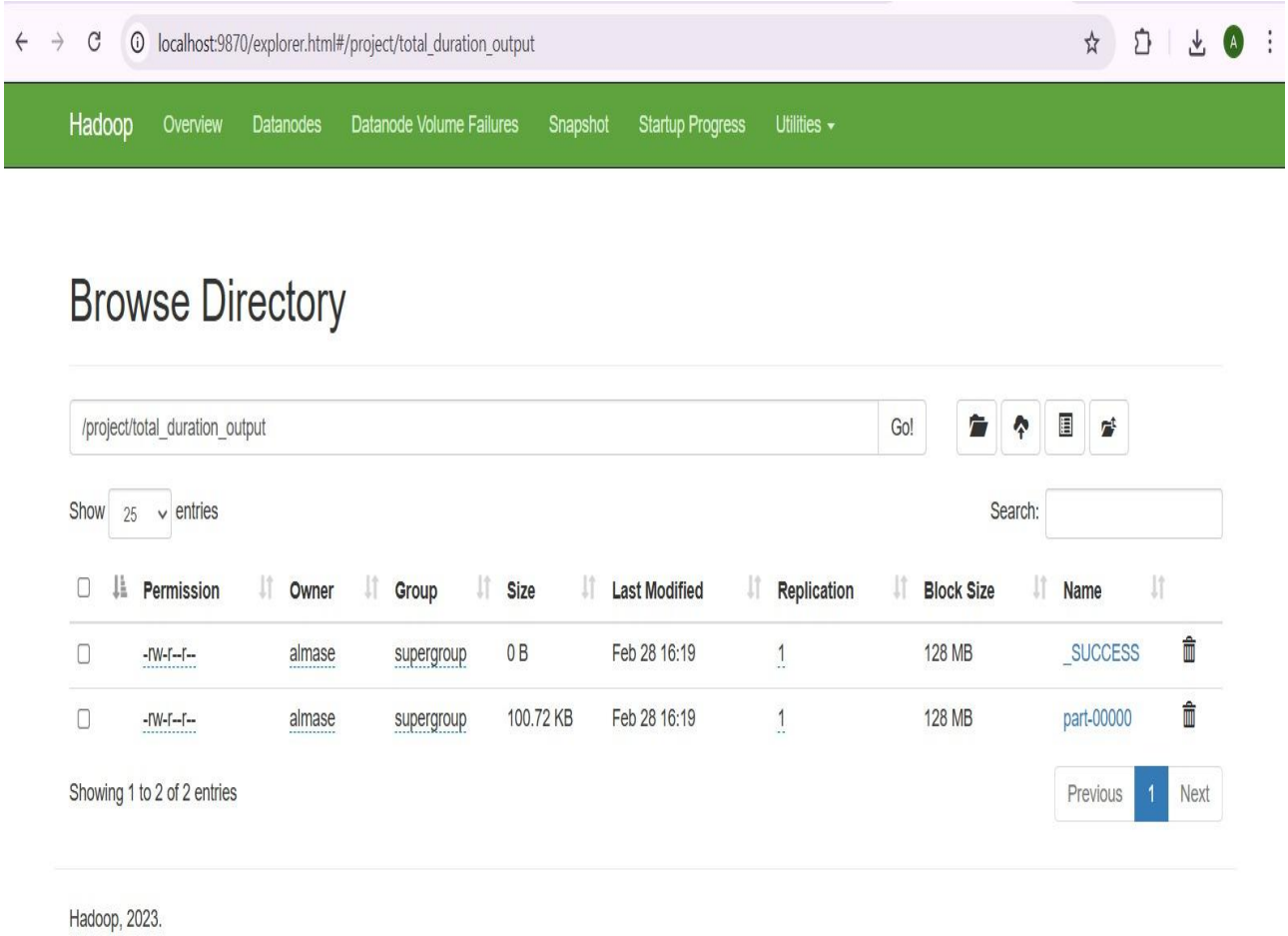
Results Screenshots

```
  GNU nano 7.2                                                    telecom_final_mapper.py *
#!/usr/bin/env python3
import sys

for line in sys.stdin:
    fields = line.strip().split(",")

    if fields[0] == "Phone Number":
        continue

    try:
        phone = fields[0]

        day_mins = float(fields[3])
        eve_mins = float(fields[6])
        night_mins = float(fields[9])
        intl_mins = float(fields[12])
        day_calls = int(fields[4])
        eve_calls = int(fields[7])
        night_calls = int(fields[10])
        intl_calls = int(fields[13])
        custserv_calls = int(fields[15])
        churn = fields[16]
        total_duration = day_mins + eve_mins + night_mins + intl_mins
        total_calls = day_calls + eve_calls + night_calls + intl_calls
        #1  Total Duration Per User
        print("UserDuration_" + phone + "\t" + str(total_duration))
        #2  Peak Usage Aggregation
        print("DayCalls\t" + str(day_calls))
        print("EveCalls\t" + str(eve_calls))
        print("NightCalls\t" + str(night_calls))
        print("IntlCalls\t" + str(intl_calls))
        #3  Churn
        if churn == "TRUE":
            print("Churn\t1")
        else:
            print("NonChurn\t1")
        #4  Fraud Detection
        if custserv_calls > 3 or intl_calls > 5:
            print("HighRisk\t1")
        #5  Heavy Users
        if total_calls > 300:
            print("HeavyUsers\t1")
    except:
        continue
```

```
  GNU nano 7.2                                                    telecom_final_reducer.py *
#!/usr/bin/env python3
import sys

current_key = None
current_total = 0.0

for line in sys.stdin:
    key, value = line.strip().split("\t")

    try:
        value = float(value)
    except:
        continue

    if current_key == key:
        current_total += value
    else:
        if current_key:
            print(current_key + "\t" + str(round(current_total,2)))
        current_key = key
        current_total = value

if current_key:
    print(current_key + "\t" + str(round(current_total,2)))
```

```
almase@LAPTOP-HMI13H4S:/mnt/c/Users/ASUS/Desktop/telecom_project$ hdfs dfs -cat /project/final_output/part-00000 | grep
UserDuration_ | sort -k2 -nr | head -10
UserDuration_778-7875   140336903.52001134
UserDuration_777-9895   117495593.16000144
UserDuration_789-7877   111744177.59999901
UserDuration_985-9789   101004096.60000177
UserDuration_766-9578   96405046.31999844
UserDuration_765-5979   92681101.92000148
UserDuration_799-9878   91597401.72000426
UserDuration_759-8999   90633810.9599873
UserDuration_798-7989   90187807.38001038
UserDuration_766-5595   90093166.02000502
almase@LAPTOP-HMI13H4S:/mnt/c/Users/ASUS/Desktop/telecom_project$ hdfs dfs -cat /project/final_output/part-00000 | grep
Calls
CustServCalls   94005715
DayCalls        9813508278
EveCalls        9812013882
IntlCalls       210727244
NightCalls      9610249953
almase@LAPTOP-HMI13H4S:/mnt/c/Users/ASUS/Desktop/telecom_project$ hdfs dfs -cat /project/final_output/part-00000 | grep
HighRisk
HighRisk        19912357
almase@LAPTOP-HMI13H4S:/mnt/c/Users/ASUS/Desktop/telecom_project$ hdfs dfs -cat /project/final_output/part-00000 | grep
HeavyUsers
HeavyUsers      31393682
almase@LAPTOP-HMI13H4S:/mnt/c/Users/ASUS/Desktop/telecom_project$ hdfs dfs -cat /project/final_output/part-00000 | grep
Churn
Churn   3823116
NonChurn        32903034
```



File information - part-00000

Download | Head the file (first 32K) | Tail the file (last 32K)

Block information – Block 0

Block ID: 1073741851

Block Pool ID: BP-821198143-127.0.1.1-1772210810028

Generation Stamp: 1027

Size: 103135

Availability:

- LAPTOP-HMI13H4S.localdomain

File contents

```
327-1058 0
327-1319 8349
327-3053 0
327-3587 13068
327-3850 0
327-3954 0
327-4795 0
327-5525 5808
```

Close

File information - part-00000

Download | Head the file (first 32K) | Tail the file (last 32K)

Block information – Block 0

Block ID: 1073741851

Block Pool ID: BP-821198143-127.0.1.1-1772210810028

Generation Stamp: 1027

Size: 103135

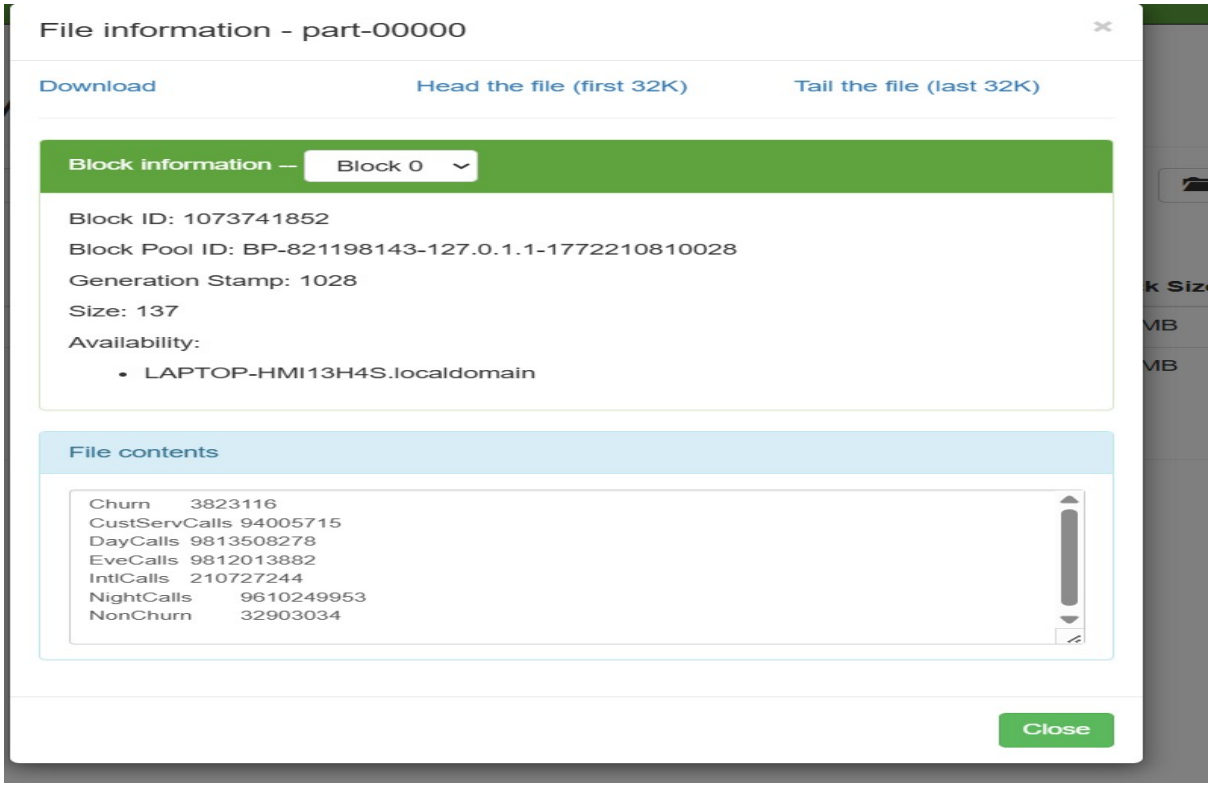
Availability:

- LAPTOP-HMI13H4S.localdomain

File contents

```
799  138666
787-9855 121605
787-9859 19965
787-9867 9438
787-9869 290763
787-9876 69696
787-9885 308187
787-9899 23958
```

Close


File information - part-00000
Download
Head the file (first 32K)
Tail the file (last 32K)
Block information — Block 0
Block ID: 1073741852
Block Pool ID: BP-821198143-127.0.1.1-1772210810028
Generation Stamp: 1028
Size: 137
Availability:
LAPTOP-HMI13H4S.localdomain
File contents
Churn 3823116
CustServCalls 94005715
DayCalls 9813508278
EveCalls 9812013882
IntlCalls 210727244
NightCalls 9610249953
NonChurn 32903034
Close